

innovate achieve lead

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Laboratory-6: Tasks

Submit at https://forms.gle/ZGMYX2TRuSEvFA3Z6

by 6:50 PM

Task 1

- For the dataset url = "https://archive.ics.uci.edu/ml/machine-learning-databases/00325/Sensorless_drive_diagnosis.txt"

```
data = pd.read_csv(url, sep=r'\s+',header=None)

X = data.iloc[:, :-1]

y = data.iloc[:, -1].astype(int)
```

- Compare Random Forest and Gradient Boosting results using 70:30 split.
- Consider n_estimators= 5, 10, 25 cases
- Compare training and test accuracy, precision, recall and F1-score
- Identify the best Random Forest model and Gradient Boosting model and plot the confusion matrix showing the 11 classes.

Task 2

- For the machine predictive maintenance dataset available in: https://archive.ics.uci.edu/ml/machine-learning-databases/00601/ai4i2020.csv
- Use features: 'Air temperature [K]', 'Process temperature [K]', 'Rotational speed [rpm]', 'Torque [Nm]', 'Tool wear [min]'
- Use 70:30 split
- Compare Decision Tree, Pre-pruned DT, Random Forest and Gradient Boosting accuracies
- Compare their precision, recall and F1-scores as well and comment on the obtained results
- Plot the confusion matrices for the best DT, pre-pruned DT, RF and GB models.